The Tracking of Morning Fatigue Status Across In-Season Training Weeks in Elite Soccer Players

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Tracking Morning Fatigue Status Across In-Season Training Weeks in Elite Soccer Players

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Purpose: To quantify the mean daily changes in training and match load and any parallel changes in indicators of morning-measured fatigue across in-season training weeks in elite soccer players. ***Methods:*** After each training session and match (TL), session ratings of perceived exertion (s-RPE) were recorded to calculate overall session load (RPE-TL) in 29 English Premier League players from the same team. Morning ratings of fatigue, sleep quality, and delayed-onset muscle soreness (DOMS), as well as submaximal exercise heart rate (HRex), postexercise heart-rate recovery (HRR%), and heart-rate variability (HRV) were recorded before match day and 1, 2, and 4 d postmatch. Data were collected for a median duration of 3 wk (range 1–13) and reduced to a typical weekly cycle including no midweek match and a weekend match day. Data were analyzed using within-subject linear mixed models. ***Results:*** RPE-TL was approximately 600 arbitrary units (AU) (95% confidence interval 546–644) higher on match day than following day ($P < .001$). RPE-TL progressively decreased by »60 AU per day over the 3 days before a match ($P < .05$). Morning-measured fatigue, sleep quality, and DOMS tracked the changes in RPE-TL, being 35–40% worse on postmatch day vs prematch day ($P < .001$). Perceived fatigue, sleep quality, and DOMS improved by 17–26% from postmatch day to 3 d postmatch, with further smaller (7%–14%) improvements occurring between 4 d postmatch and prematch day ($P < .01$). There were no substantial or statistically significant changes in HRex, HRR%, or HRV over the weekly cycle ($P > .05$). ***Conclusions:*** Morning-measured ratings of fatigue, sleep quality, and DOMS are clearly more sensitive than HR-derived indices to the daily fluctuations in session load experienced by elite soccer players in a standard in-season week.

Keywords: training, performance, wellness, recovery

It is important to allow sufficient recovery between training sessions and competitions. An imbalance between training/competition load and recovery may, over extended periods of time, contribute to potentially long-term debilitating effects associated with overtraining.[1] Consequently, attention is increasingly being given to the evaluation of monitoring tools that may indicate the general fatigue status of athletes. These indicators include heart-rate (HR) -derived indices,[2] salivary hormones, neuromuscular indices,[3] and perceived wellness ratings.[4–6]

A valid marker of fatigue should be sensitive to variability in training load.[7] Researchers have, therefore, examined the sensitivity of potential measures of fatigue to daily fluctuations in training load in elite team sport athletes.[4–6] For example, in both elite Australian Rules Football[4] and elite soccer[6] players, small to large and statistically significant correlations were reported between fluctuations in daily training load and changes in both perceived ratings of wellness and vagal-related HR variability indices. These findings suggest that such measures show particular promise as acute, simple, noninvasive assessments of fatigue status in elite team-sport athletes.

Further evaluation of the validity of potential fatigue measures can be undertaken by examining their sensitivity to prescribed changes in training load over extended periods of time. Although these relationships have been examined in individual endurance-based sports,[8,9] limited attention has been given to elite team sport athletes,[5] who are required to compete weekly and often biweekly across the competition period. In these athletes, a key component of the in-season and within-week training prescription resides around the need to periodize the training load to minimize player fatigue ahead of the weekly matches.[10] Gastin et al[5] recently reported that subjective ratings of physical and psychological wellness (fatigue, muscle strain, hamstring strain, quadriceps strain, pain/stiffness, power, sleep quality, stress, and well-being) were sensitive to within-week training manipulations (ie, improved steadily throughout the week to a game day low) in elite Australian Football League players. However, to our knowledge, no researcher has examined the sensitivity of simple, noninvasive potential measures of fatigue across in-season training weeks in elite soccer players. Because differences exist in the physiological demands between team sports, it is important to determine which potential fatigue variables are sensitive to changes in load associated with specific sports. Therefore, our aim was to quantify any changes in perceived ratings of wellness and objective measures of vagal-related HR indices that occur across standard in-season training weeks in elite soccer players.

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Methods

Subjects

Twenty-nine soccer players (age 27 ± 5.1 y, height 181 ± 7.1 cm, weight 78 ± 6.1 kg) from the same team competing in the English Premier League participated in this study.

Design

Player training load was assessed on 6 days: on the prematch day, match day, and 1, 2, 4, and 5 days after the match across standard training weeks (no midweek match; median of 3 wk/player; range 1–13) during the 2012–13 in-season competitive period (August–May). Players were required to complete a minimum match duration of 60 minutes in order for their weekly data to be included in the current study. Players did not train and were given a day off 3 days after a match. Players took part in normal team training throughout the period as prescribed by the coaching staff. Players performed a range of recovery interventions the day following the match, including low-intensity cycling, foam rolling, and hydrotherapy. All players were fully familiarized with the assessments in the weeks before completion of the main experimental trials.

Fatigue measures were assessed on the day before the match and 1, 2, and 4 days after the match. On the day of the fatigue assessments (perceived ratings of wellness, submaximal HR, HR recovery [HRR], and HR variability [HRV]), players arrived at the training ground laboratory after having refrained from caffeine and alcohol intake for at least 12 hours before each assessment point. Fatigue measures were subsequently taken before the players commencing normal training. All trials were conducted at the same time of the day to avoid the circadian variation in body temperature.[11] Players were not allowed to consume fluid at any time during the fatigue assessments. The study was approved by Liverpool John Moores University ethics committee. All players provided written informed consent. Before inclusion into the study, players were examined by the club physician and were deemed to be free from illness and injury.

Methods

Training Load Assessment. Individual player daily training load was monitored throughout the assessment period. Load (RPE-TL, in arbitrary units [AU]) was estimated for all players by multiplying total training or match session (TL) duration (min) with session ratings of perceived exertion (RPE).[12] Despite several influences, the usefulness of RPE in elite soccer has previously been observed.[13] Player RPE was collected within 20 to 30 minutes after cessation of the training session/match.[13]

Perceived Ratings of Wellness. A psychometric questionnaire was used to assess general indicators of player wellness.[6,14] The questionnaire comprised 3 questions relating to perceived overall fatigue, sleep quality, and delayed-onset muscle soreness (DOMS)[6,14] Each question was scored on a 7-point scale (scores of 1–7, with 1 and 7 representing very, very poor [negative state of wellness] and very, very good [positive state of wellness], respectively). Coefficients of variation for the 3 indices ranged from 9% to 13%.[6]

Submaximal Exercise HR, Postexercise HRR, and HRV. Players completed an indoor submaximal 5-minute cycling/5-minute recovery test (Keiser, Fresno, CA, USA) before commencing every session.[2,6] All players were tested together at a fixed exercise intensity of 130 W (85 rpm). The present intensity was selected to minimize anaerobic energy contribution and to permit a rapid return of HR to baseline for short-term HRV measurements.[15] On completion of exercise, the players remained seated in silence for 5 minutes.

Submaximal exercise HR (HRex) was calculated using the average of the final 30 seconds of the cycle test.[16] HRR expressed as the absolute (HRR) and relative (%HRR) change in HR between the final 30 seconds (average) of the 5-minute cycling test and 60 seconds after cessation of exercise were calculated as previously described.[15,17] The coefficients of variation for HRex, HRR, and %HRR were 3%, 14% and 10%, respectively.[6] HRV was measured during the recovery period and expressed as the natural logarithm of the square root of the mean of the sum of squares of differences between adjacent normal R-R intervals (Ln rMSSD) as previously described[15] using Polar software (Polar Precision Performance SW 5.20, Polar Electro, Kempele, Finland). Ln rMSSD has previously been shown to have greater reliability and validity than spectral indices of HRV over short assessment periods.[18,19] The coefficient of variation for Ln rMSSD was 10%.[6]

Statistical Analyses

It was assumed that if an indicator of fatigue was not, at the very least, sensitive to differences between the different loads on prematch day and the postmatch day, it could not be considered useful. Therefore, for the purpose of our sample size estimation, our primary comparison was between the prematch and postmatch days. In a previous study, coefficients of variation of approximately 10% were reported for the indicators of fatigue we studied.[6] Using this information of within-subjects variability, we estimated that a sample size of 29 would allow the detection of a difference in fatigue between prematch and postmatch days of approximately 9% (2-tailed paired t test, 90% statistical power, $P < .05$).

A within-subjects linear mixed model was used to quantify mean differences between days along with the respective 95% confidence intervals (CIs). It is difficult to ascertain the exact relative influence of each study outcome on the actual performance of a soccer team, for example, the standard effect size (ES) of a particular outcome may be high in response to training or an intervention, but the relative influence of this outcome on actual soccer performance may be low.[20] Nevertheless, standardized ESs, estimated from the ratio of the mean difference to the pooled SD, were also calculated for each study outcome and interpreted in the Discussion section. ES values of 0.2, 0.5, and 0.8 were considered to represent small, moderate, and large differences, respectively.[21] When the model residuals were skewed or heteroscedastic, data were log-transformed and reanalyzed. We adopted the least significant difference approach to multiple comparisons in line with the advice in Perneger[22] and Bland and Altman.[23]

Results

Training Load

The RPE-TL was greatest on match day (»600 AU). The peak-trough difference in RPE-TL was approximately 550 AU (95% CI, 546–644 AU) between match day and the following day ($P < .001$). The RPE-TL progressively decreased by ~60 AU/d over the 3 days before a match ($P < .05$) (Figure 1).

Perceived Ratings of Wellness

All the wellness outcomes showed a 35% to 40% worsening on the postmatch day versus the prematch day. The 95% CIs for these changes were 1.2 to 1.6 AU, 1.0 to 1.5 AU, and 1.1 to 1.5 AU for perceived fatigue, sleep quality, and DOMS, respectively ($P < .001$). Wellness outcomes then improved by 17% to 26% between postmatch day and 2 days postmatch. The 95% CIs for these changes were 0.7 to 1.1 AU, 0.7 to 1.2 AU, and 0.4 to 0.9 AU for perceived fatigue, sleep quality, and DOMS, respectively. Wellness ratings

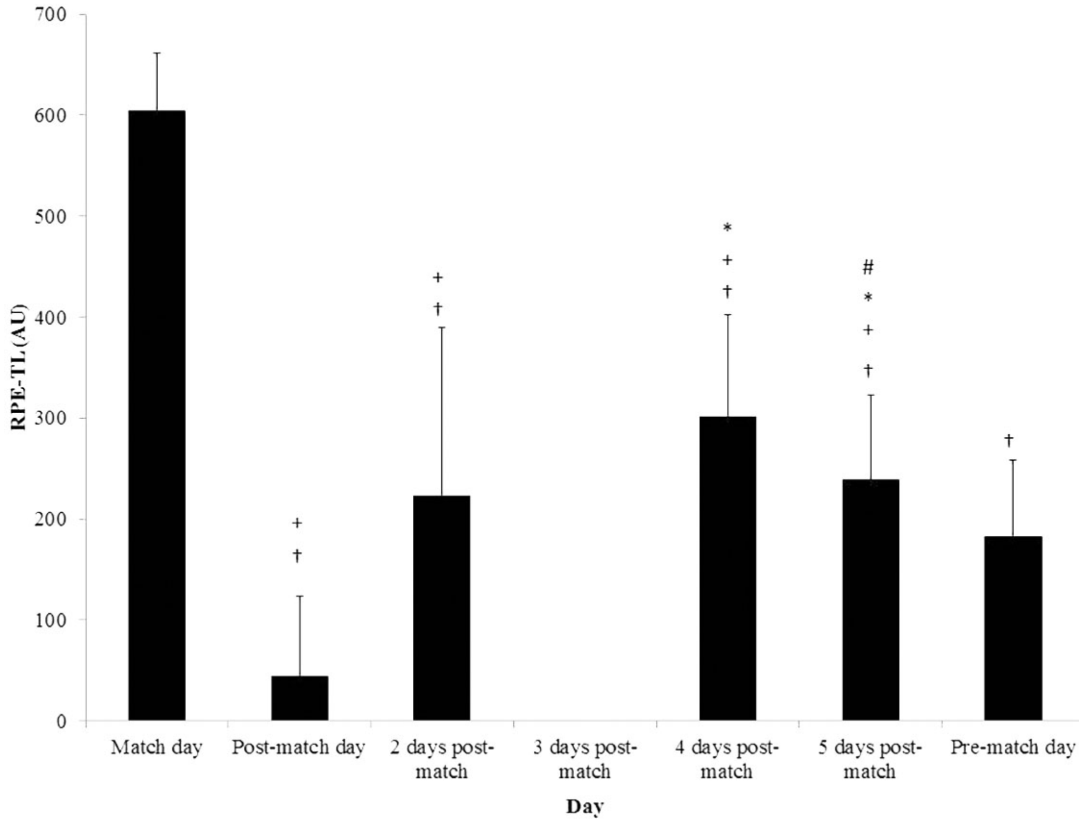

Figure 1 — Training load (in arbitrary units [AU]) across in-season training weeks (mean ± SD). †Significant difference vs match day. +Significant difference vs prematch day. *Significant difference vs 2 days postmatch. #Significant difference vs 4 days postmatch. Abbreviation: RPE-TL, session ratings of perceived exertion multiplied by total training/match time.

then remained relatively stable between the second and fourth day postmatch. Further smaller (7%–14%) improvements occurred between the fourth day postmatch and prematch day ($P < .01$). The 95% CIs for these changes were 0.2 to 0.6 AU, 0.1 to 0.6 AU, and 0.4 to 0.7 AU for perceived fatigue, sleep quality, and DOMS (Table 1).

HR Indices

There were no substantial or statistically significant changes in HRex, HRR%, and HRV over all the weekdays ($P > .05$) (Table 1).

Discussion

The aim of the current study was to quantify the mean daily changes in training load and parallel changes in potential fatigue measures across in-season training weeks in elite soccer players. The main finding was that perceived ratings of wellness but not HR-derived indices are sensitive to the fluctuations in training load experienced by elite soccer players across in-season training weeks that involve only 1 match per week (no midweek match).

Elite soccer players are required to compete on a weekly and often biweekly (midweek game) basis with additional training administered in-between matches. Training load prescribed by coaches should, therefore, serve to ensure that fatigue is reduced on the days when players are engaged in competition. In the current study, only training weeks containing no midweek game were used to examine changes in fatigue across a "standard" training week. A clear attempt to periodize training load across the week was observed, with the lowest load prescribed the day following a match with large (ES >1.3) and statistically significant increases in training load prescribed 2 and 4 days after the match. During the 2 subsequent days (fifth day postmatch and prematch day) there was a moderate (ES = 0.7) and statistically significant reduction in training load in the lead into the next game (Figure 1). So far, little information currently exists with regard to the patterns of training load undertaken by elite soccer players.[10,24] Interestingly, the pattern of training load exhibited in the current study differs from that seen in recent observations in Premier League players where only a reduction in daily training load was observed 1 day before a match compared with the other training days.[10] However, Malone et al[10] analyzed all training weeks throughout the in-season competition period, including those containing a midweek game. The combination of this and dissimilarities in coaching philosophy and training methodology likely explain the difference in training load periodization to the current study. Further research is warranted to explore the patterns of training load experienced by elite players across different phases of the season.

Perceived ratings of wellness represent an increasingly popular method to assess athlete fatigue. Recent work in both elite soccer[6] and Australian Rules Football[4] players demonstrated that such ratings are sensitive to daily fluctuations in training load. Further information concerning the validity of potential markers of fatigue can be derived by examining their sensitivity to prescribed changes in training load over extended periods of time. Although these relationships have been examined in individual endurance-based sports[8,9] a limited attempt, to date, has been made to determine the

Table 1 Perceived Ratings of Fatigue, Sleep Quality, and Delayed-Onset Muscle Soreness (DOMS) and HRex, HRR, and Ln rMSSD Across In-Season Training Weeks (Mean ± SD)

Fatigue measure	Postmatch day	2 d postmatch	4 d postmatch	Prematch day
Fatigue (AU)	3.4 ± 0.6[a]	4.4 ± 0.7[a,b]	4.5 ± 0.7[a,b]	5.0 ± 0.6
Sleep quality (AU)	3.9 ± 1.2[a]	4.8 ± 0.9[a,b]	4.7 ± 1.0[a]	5.2 ± 0.8
DOMS (AU)	3.6 ± 0.6[a]	4.3 ± 0.7[a,b]	4.4 ± 0.7[a,b]	5.1 ± 0.8
HRex (beats/min)	119 ± 13	117 ± 14	119 ± 15	118 ± 13
HRR (%)	72.1 ± 7.7	71.5 ± 7.5	70.2 ± 7.7	70.9 ± 7.1
Ln rMSSD (ms)	3.31 ± 0.71	3.44 ± 0.69	3.28 ± 0.76	3.33 ± 0.64

Abbreviations: AU, arbitrary units; HRex, exercise heart rate; HRR, heart rate recovery; Ln rMSSD, natural logarithm of the square root of the mean of the sum of squares of differences between adjacent normal R-R intervals.

Note: Scores of 1–7 with 1 and 7 representing very, very poor (negative state of wellness) and very, very good (positive state of wellness), respectively.

[a] Significant difference vs prematch day. [b] Significant difference vs postmatch day.

sensitivity of tools for monitoring fatigue over extended periods of time in elite team-sport players.[5] In the current study, the between-days changes in perceived wellness across the weekly training cycle closely reflected the prescribed distribution of training load. Moderate to large (ES 0.5–2.4) statistically significant changes (35–40%) in perceived ratings of fatigue, sleep quality, and DOMS were observed across the training week with the greatest and least amount of perceived fatigue, sleep quality, and DOMS reported on the day following and the day before a match, respectively (Table 1). These observations are consistent with findings in Australian Football where perceived ratings of fatigue, muscle strain, hamstring strain, quadriceps strain, pain/stiffness, power, sleep quality, stress, and well-being improved by ~30% throughout the week.[5] Interestingly, previous work in elite soccer examining the sensitivity of perceived ratings of DOMS and sleep quality to acute daily fluctuations in training load failed to observe any association.[6] However, any effect of training and match load on DOMS and sleep quality, in particular the effects of match play, may materialize over a number of days rather than immediately after the session.[25] Collectively, previous observations examining daily sensitivity in elite soccer[6] and Australian Rules players[4,5] combined with the present findings suggest that attempts to fully examine the sensitivity of potential markers of fatigue to changes in training and match load should be undertaken over both acute (daily) and extended periods of time.

The increased perception of fatigue, poor sleep, and DOMS currently observed following match play is consistent with changes in biochemical status and reduced physical and neuromuscular performance observed in the hours and days following soccer competition.[24,26–29] In contrast to many of the latter assessments, perceived wellness scales represent a valid, time-efficient, and noninvasive means through which to derive information pertaining to a player's fatigue status. Such characteristics are important during the in-season competitive phase, particularly during periods when players are required to compete in 2 or 3 matches over a 7-day period where time constraints may restrict the use of more invasive tests, and maximal performance tests may further debilitate the physical status of players and/or increase the risk of injury.[30] In the current study, perceived ratings of fatigue and DOMS remained similar over the second and fourth day postmatch despite a rest day 3 days after a match. This plateau may be due to the magnitude of the training load assigned 2 days postmatch (224 ± 166 AU), which provided sufficient stimulus to blunt a linear improvement in player fatigue/recovery 4 days postmatch and/or the fact that players were relatively well recovered 2 days postmatch (fatigue 4.6 AU, sleep quality 4.8 AU, DOMS 4.3 AU). Interestingly, the progressive reduction in training load during the 3 days leading into a match was accompanied by further moderate-to-large (ES 0.6–0.9) statistical significant improvements in perceived wellness the day before a match (fatigue 5.0 AU, sleep quality 5.1 AU, DOMS 5.2 AU), which suggests the players were still not fully recovered 4 days postmatch (Table 1). The time required to fully recover following match play has been shown to vary markedly (24–72 h) depending on the nature of the physiological parameter assessed.[25] Furthermore, the rate of recovery is likely to be influenced by a myriad of factors including the inherent variability in match demands[31] and the athlete's level of fitness.[32] Alongside the changes in perceived ratings of fatigue and DOMS, moderate to large (ES 0.6–1.4) statistical significant changes in ratings of sleep quality were also observed across the week, with the highest and lowest levels of sleep quality observed during the evening of the fifth day postmatch and the evening immediately after a match, respectively. These changes indicate the severe debilitating effects of the match on perceived ratings of sleep quality. Indeed DOMS, inflammation, nervous system activity, and central excitation have all been reported as potential mechanisms of poor sleep after competition.[33] Interestingly, data from elite endurance athletes have frequently shown reductions in sleep quantity the night before competition.[34] Perceived ratings of sleep were not measured the night before matches in the current study, and, therefore, a reduction may have occurred. Future work is required to further understand the effects of training and match load on perceived and objective measures of sleep quality.

HR indices (HRex, HRV, and HRR) have recently been proposed as a noninvasive method to measure variations in the autonomic nervous system in attempt to understand athlete adaptation/fatigue status.[2] The use of vagal-related time domain indices such as Ln rMSSD compared with spectral indices of HRV have been found to have greater reliability and are ideal for assessments undertaken over shorter periods of time.[18,19] Recent work in elite soccer players observed small ($r = .2$), significant correlations between daily fluctuations in training load and Ln rMSSD.[6] Similarly, in Australian Rules Football players undergoing preseason training, very large ($r = .80$) and moderate ($r = -.40$) significant correlations were observed between daily training load and submaximal HRex and a vagal-related index of HRV (LnSD1).[4] In the current study, HRV (Ln rMSSD), HRex, and HRR (%) remained unchanged across the training week (Table 1) despite the large statistical significant fluctuations in training load (Figure 1). This suggests that in contrast to perceived ratings of wellness, such indices lack the sensitivity to provide information concerning changes in the fatigue status of elite soccer players across in-season training weeks. It should be noted that the average daily training load in the current study (RPE-TL 228) is considerably lower than that reported during an AFL preseason training camp (RPE-TL 746) where daily readings of HRex and HRV were negatively and positively associated with load, respectively. It is, therefore, possible that the magnitude of the fluctuations in daily training load across the training week in the current study were insufficient to elicit changes in HRex and HRV.[4] Alternatively, the shorter 7-day period over which observations were made in the current study may not have been sufficient to detect any physiological change. Indeed data derived from elite triathletes and adolescent handball players suggests the use of a single data point could be misleading for practitioners due to the high day-to-day variation in these indices.[35,36] In elite triathletes, when data were averaged over a week or using a 7-day rolling average significant large correlations were found with 10-km running performance compared with a single assessment point where negligible relationships were seen.[35] In addition, changes in monthly HRV measurements were not sensitive to changes in performance indices in young handball players.[36] Future research is needed to establish whether sensitivity of HR indices are seen over longer training periods in elite soccer players.

Conclusions

Perceived ratings of wellness are clearly more sensitive than HR-derived indices to the within-week fluctuations in training load experienced by elite soccer players during typical in-season training weeks. Therefore, perceived ratings of wellness show particular promise as simple, noninvasive assessments of fatigue status in elite soccer players throughout typical in-season weeks in elite soccer players.

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